



QUALITY IN SPORT

eISSN 2450-3118 · Open Access · Peer-reviewed

apcz.umk.pl/QS

Nicolaus

Copernicus

University in Toruń



OBERSZTYN, Julia, MŁICZEK, Aleksandra, SZCZECIŃSKA, Adrianna, ROCZYŃSKI, Brajan, POŁOK, Izabela, and ŁOMPIEŚ, Kinga. Hypercholesterolemia in Underweight Individuals: Mechanisms, Clinical Significance and Practical Recommendations. *Quality in Sport*. 2026;73:75089. <https://doi.org/10.12775/QS.2026.73.75089>

ARTICLE TIMELINE

Received: 06.08.2026. Revised: 26.08.2026. Accepted: 06.09.2026. Published: 17.09.2026.

The journal has been awarded 20 points in the parametric evaluation by the Polish Ministry of Higher Education and Science (Annex to the announcement of 05.01.2024, No. 32553). Unique Journal Identifier: 201398. Scientific disciplines assigned: Economics and Finance (Field of Social Sciences); Management and Quality Sciences (Field of Social Sciences).

Punkty Ministerialne z 2019 – aktualny rok 20 punktów. Załącznik do komunikatu Ministra Szkolnictwa Wyższego i Nauki z dnia 05.01.2024 Lp. 32553. Posiada Unikatowy Identyfikator Czasopisma: 201398. Przypisane dyscypliny naukowe: Ekonomia i finanse (Dziedzina nauk społecznych); Nauki o zarządzaniu i jakości (Dziedzina nauk społecznych). © The Authors 2026.

OPEN ACCESS · CC BY-NC-SA 4.0 This article is published with open access under the License Open Journal Systems of Nicolaus Copernicus University in Toruń, Poland, and is distributed under the terms of the Creative Commons Attribution Non-commercial Share Alike License (<http://creativecommons.org/licenses/by-nc-sa/4.0/>), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the work is properly cited. The authors declare no conflict of interest regarding the publication of this paper.

Hypercholesterolemia in Underweight Individuals: Mechanisms, Clinical Significance and Practical Recommendations

Julia Obersztyn

ORCID: <https://orcid.org/0009-0001-7547-4556>

E-mail: julia.ob@op.pl

University Clinical Hospital in Poznań

Ul. Przybyszewskiego 49, 60-355, Poznań, Poland

Aleksandra Mliczek

ORCID: <https://orcid.org/0009-0000-9883-3307>

E-mail: al.mliczek@gmail.com

University Clinical Hospital in Poznań

Ul. Przybyszewskiego 49, 60-355, Poznań, Poland

Adrianna Szczecińska

ORCID: <https://orcid.org/0009-0004-1714-0832>

E-mail: adrianna915@gmail.com

University Clinical Hospital in Poznań

Ul. Przybyszewskiego 49, 60-355, Poznań, Poland

Brajan Roczyński

ORCID: <https://orcid.org/0009-0003-9705-4557>

E-mail: brajan.roczynski@wp.pl

University Clinical Hospital in Poznań

Ul. Przybyszewskiego 49, 60-355, Poznań, Poland

Izabela Polok

ORCID: <https://orcid.org/0009-0005-6029-165X>

E-mail: izapolok999@gmail.com

University Clinical Hospital in Poznań

Ul. Przybyszewskiego 49, 60-355, Poznań, Poland

Kinga Łompiś

ORCID: <https://orcid.org/0009-0007-7402-5146>

E-mail: kinga.lo@op.pl

Saint Adalbert Hospital in Gdańsk

Aleja Jana Pawła II 50, 80-462, Gdańsk, Poland

Corresponding author:

Julia Obersztyn, E-mail: julia.ob@op.pl

Abstract

Introduction. Hypercholesterolemia is typically associated with obesity and metabolic syndrome, yet elevated total and LDL cholesterol are also frequently observed in underweight individuals, including patients with anorexia nervosa (AN). This paradox can cause diagnostic uncertainty when lipid results are interpreted based on concepts derived from obesity-related dyslipidemia.

Aim. To summarize current evidence on mechanisms underlying hypercholesterolemia in underweight individuals, with emphasis on AN, and to outline its clinical significance and management.

Materials and Methods. A narrative review of PubMed and Google Scholar literature published up to July 2026, prioritizing systematic reviews, meta-analyses, and observational studies on lipid abnormalities in underweight populations.

Results. Hypercholesterolemia in starvation states arises from reduced hepatic LDL-receptor activity, a low-T3 endocrine state, altered cortisol, insulin and leptin signalling, impaired bile acid synthesis, and altered lipoprotein transport, rather than dietary cholesterol intake. These adaptations differ from obesity-driven dyslipidemia and generally improve with nutritional rehabilitation.

Conclusions. Hypercholesterolemia associated with underweight is usually a reversible metabolic adaptation to prolonged energy deficiency. Nutritional rehabilitation should be the primary treatment strategy, whereas persistent lipid abnormalities after weight restoration require further diagnostic evaluation.

Keywords: anorexia nervosa, hypercholesterolemia, underweight, malnutrition, lipid metabolism, nutritional rehabilitation

1. Introduction

Hypercholesterolemia is commonly associated with obesity, metabolic syndrome, and increased cardiovascular risk. In clinical practice, elevated total cholesterol (TC) and low-density lipoprotein cholesterol (LDL-C) concentrations are therefore often interpreted as markers of a metabolic profile requiring lifestyle modification or lipid-lowering therapy. However, elevated cholesterol levels may also occur in individuals with low body weight, including patients with anorexia nervosa (AN), chronic malnutrition, or prolonged energy deficiency. This apparent paradox may lead to diagnostic uncertainty and inappropriate clinical management [1–3].

Among underweight individuals, anorexia nervosa is the best-studied condition associated with hypercholesterolemia. A recent systematic review and meta-analysis demonstrated that patients with acute AN have significantly higher concentrations of total cholesterol, LDL-C, high-density lipoprotein cholesterol (HDL-C), triglycerides, and apolipoprotein B than healthy controls [1]. Although lipid concentrations often improve after nutritional rehabilitation, normalisation is not immediate and the mechanisms responsible for these abnormalities remain incompletely understood [1–3]. Proposed explanations include reduced LDL receptor activity, altered hepatic cholesterol metabolism, hormonal adaptations to starvation, and changes in cholesterol transport [2,3].

Similar lipid abnormalities have also been described in other conditions characterised by prolonged low energy availability, including severe malnutrition and restrictive eating behaviors. Therefore, elevated cholesterol concentrations in underweight individuals should not automatically be interpreted as evidence of primary dyslipidemia. Instead, clinicians should distinguish adaptive metabolic changes associated with starvation from disorders requiring further investigation, such as familial hypercholesterolemia or endocrine disease [1,2].

The aim of this narrative review is to summarize current evidence on the mechanisms responsible for hypercholesterolemia in underweight individuals, with particular emphasis on anorexia nervosa and other conditions associated with low energy availability. The review also discusses the clinical significance of these lipid abnormalities and provides practical recommendations for their diagnosis and management.

2. Materials and methods

This article is a narrative review. A literature search was conducted using the PubMed (MEDLINE) and Google Scholar databases for studies published in English up to July 2026. Search terms included combinations of "hypercholesterolemia", "cholesterol", "low body

weight", "underweight", "anorexia nervosa", "malnutrition", "starvation", "energy deficiency", "RED-S", "lipid metabolism", "LDL cholesterol", and "refeeding". Priority was given to systematic reviews, meta-analyses, randomized and observational studies investigating lipid abnormalities in underweight individuals and the mechanisms responsible for these changes. Reference lists of relevant publications were manually screened to identify additional studies. Case reports, conference abstracts without peer-reviewed full texts, and non-English publications were excluded.

3. Cholesterol metabolism during starvation

3.1. Normal cholesterol metabolism

Cholesterol is an essential component of cell membranes and serves as a precursor for steroid hormones, bile acids, and vitamin D. It is obtained from dietary sources and endogenous synthesis, which occurs mainly in the liver. Cholesterol is transported in the circulation by lipoproteins. Low-density lipoprotein (LDL) delivers cholesterol from the liver to peripheral tissues, whereas high-density lipoprotein (HDL) participates in reverse cholesterol transport by carrying excess cholesterol back to the liver for excretion [4].

Under physiological conditions, cholesterol homeostasis is maintained by a balance between hepatic cholesterol synthesis, intestinal absorption, LDL receptor-mediated clearance, and biliary cholesterol excretion. When circulating LDL cholesterol increases, hepatic LDL receptor expression normally increases, enhancing LDL clearance and preventing excessive cholesterol accumulation. At the same time, endogenous cholesterol synthesis decreases through inhibition of 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase, the rate-limiting enzyme of cholesterol synthesis [4,5].

During prolonged starvation, these regulatory mechanisms become altered. Although dietary cholesterol intake is markedly reduced, serum cholesterol concentrations may remain unchanged or even increase. This paradox suggests that lipid abnormalities in underweight individuals are primarily caused by metabolic adaptations rather than excessive dietary cholesterol intake [1,4].

3.2. Metabolic adaptations to starvation

Starvation induces profound metabolic adaptations that allow the body to conserve energy and maintain the function of vital organs. Glycogen stores become depleted within the first 24 hours of fasting, after which energy production gradually shifts toward lipolysis, fatty acid oxidation, and ketone body production. At the same time, resting energy expenditure decreases and

several endocrine adaptations develop, including lower triiodothyronine (T3), insulin, and leptin concentrations, together with increased cortisol secretion [1,6,7].

These hormonal changes influence lipid metabolism. Reduced thyroid hormone activity slows hepatic LDL receptor expression and decreases cholesterol clearance, while lower metabolic rate reduces cholesterol turnover. In addition, prolonged energy deficiency modifies bile acid synthesis and cholesterol excretion. Rather than representing pathological dyslipidemia, these changes are generally considered as adaptive responses that preserve essential lipid substrates during periods of nutrient deprivation [1,2,6].

Importantly, these metabolic adaptations differ from those observed in obesity-related hypercholesterolemia. In obesity, elevated LDL cholesterol is mainly associated with insulin resistance, increased hepatic lipoprotein production, and chronic low-grade inflammation [8]. In contrast, hypercholesterolemia in underweight individuals develops despite low fat intake, reduced adipose tissue, and increased insulin sensitivity, suggesting a fundamentally different pathophysiological mechanism [1].

4. Mechanisms of hypercholesterolemia in underweight individuals

4.1. Reduced LDL receptor activity

Among the proposed mechanisms, reduced hepatic LDL receptor activity has received the greatest attention. Under normal physiological conditions, LDL receptors located on hepatocytes remove circulating LDL particles from the bloodstream and maintain cholesterol homeostasis. Impaired receptor activity decreases LDL clearance, resulting in elevated plasma LDL cholesterol concentrations [9,10].

Several studies suggest that starvation and anorexia nervosa are associated with reduced LDL receptor expression or activity. Decreased triiodothyronine (T3) concentrations, a characteristic endocrine adaptation to prolonged energy deficiency, appear to play an important role. Thyroid hormones normally stimulate hepatic LDL receptor synthesis; therefore, the low-T3 state observed in anorexia nervosa may reduce LDL uptake and contribute to hypercholesterolemia [10,11].

Evidence supporting this mechanism also comes from studies of nutritional rehabilitation. Serum LDL cholesterol often decreases during weight restoration despite increased dietary cholesterol intake, suggesting that recovery of hepatic LDL receptor function is more important than dietary cholesterol intake [1,10]. These observations indicate that hypercholesterolemia in underweight individuals is largely a consequence of altered cholesterol clearance rather than excessive cholesterol production.

Nevertheless, reduced LDL receptor activity alone does not fully explain the lipid abnormalities observed in all patients. Some individuals maintain elevated HDL cholesterol or total cholesterol even after partial weight recovery, while others show only modest lipid changes despite severe malnutrition [1]. These findings suggest that additional metabolic pathways contribute to the development of hypercholesterolemia.

4.2. Hormonal adaptations to starvation

Prolonged energy deficiency induces profound endocrine changes that help preserve energy but also influence lipid metabolism. One of the most consistent findings in anorexia nervosa is the development of the so-called low-T3 syndrome, characterised by reduced circulating triiodothyronine concentrations despite normal or slightly reduced thyroxine (T4) levels. This adaptation lowers resting energy expenditure and reduces metabolic rate, but it may also decrease hepatic LDL receptor expression and slow cholesterol clearance [11,12].

Starvation is also associated with reduced insulin and leptin concentrations and increased cortisol secretion. Low insulin suppresses anabolic pathways, whereas leptin deficiency reflects depleted fat stores and contributes to neuroendocrine adaptations. Elevated cortisol promotes lipolysis and mobilization of fatty acids, altering hepatic lipid metabolism. Together, these hormonal changes modify cholesterol synthesis, transport, and clearance rather than simply increasing cholesterol production [11–14].

Although each endocrine alteration probably has only a modest effect, their combined influence may explain why hypercholesterolemia develops despite minimal dietary cholesterol intake. Importantly, many of these hormonal abnormalities improve during nutritional rehabilitation, paralleling normalisation of the lipid profile in many patients [1,11].

4.3. Altered hepatic cholesterol metabolism and bile acid synthesis

The liver plays a central role in maintaining cholesterol homeostasis through cholesterol synthesis, lipoprotein production, and conversion of cholesterol into bile acids. During prolonged starvation, several of these processes are altered as part of the metabolic adaptation to energy deficiency. Rather than reflecting excessive cholesterol production, hypercholesterolemia appears to result from reduced cholesterol turnover and impaired clearance [1,15].

One proposed mechanism is a reduction in bile acid synthesis. Because bile acids represent the main route of cholesterol elimination, decreased conversion of cholesterol into bile acids may contribute to cholesterol accumulation in the circulation. Evidence from metabolic studies in

anorexia nervosa suggests that patients with the restrictive subtype have lower bile acid synthesis than healthy individuals, supporting the concept that impaired cholesterol catabolism contributes to elevated serum cholesterol concentrations [15].

These findings are consistent with the broader concept that starvation slows many energy-consuming metabolic pathways. Reduced hepatic cholesterol turnover, together with decreased LDL clearance, probably contributes to the persistence of elevated serum cholesterol despite very low dietary cholesterol intake. However, current evidence suggests that this mechanism acts together with other metabolic adaptations rather than representing the sole explanation for hypercholesterolemia.

4.4. Cholesterol transport and CETP activity

Changes in cholesterol transport may also contribute to hypercholesterolemia in underweight individuals. Cholesteryl ester transfer protein (CETP) facilitates the exchange of cholesteryl esters and triglycerides between HDL and apoB-containing lipoproteins and therefore plays an important role in lipoprotein remodeling. Increased CETP activity has been reported in patients with anorexia nervosa and has been proposed as an adaptation that accelerates cholesterol turnover under conditions of low dietary intake [11].

In addition to increased CETP activity, several studies have demonstrated higher circulating concentrations of other apolipoproteins A-I, B, C-II, C-III, and E in patients with acute anorexia nervosa. These findings suggest that starvation affects not only cholesterol concentrations but also the composition and transport of circulating lipoproteins [1].

Despite these observations, it remains uncertain whether increased CETP activity represents a primary mechanism or a secondary adaptation to starvation, and how it interacts with the concurrent changes in receptor-mediated clearance and hepatic lipoprotein production described above. Current evidence indicates that alterations in cholesterol transport are unlikely to explain hypercholesterolemia independently but probably amplify the effects of reduced LDL clearance and altered hepatic cholesterol metabolism. Therefore, CETP should be considered one component of a broader adaptive response rather than the principal cause of lipid abnormalities in underweight individuals.

5. Clinical significance of hypercholesterolemia in underweight individuals

5.1. Is hypercholesterolemia harmful in underweight patients?

Although elevated LDL cholesterol is a well-established cardiovascular risk factor in the general population, its clinical significance in underweight individuals remains uncertain.

Hypercholesterolemia associated with anorexia nervosa or prolonged starvation develops in a metabolic environment that differs markedly from obesity-related dyslipidemia. Patients are typically young, insulin sensitive, normotensive, and have very low visceral fat mass. Therefore, elevated cholesterol concentrations in this setting may not carry the same cardiovascular implications as primary hypercholesterolemia [16–18].

Several studies have shown that lipid abnormalities often improve during nutritional rehabilitation without lipid-lowering treatment, suggesting that hypercholesterolemia is largely a reversible metabolic adaptation rather than a permanent disorder of lipid metabolism [1,16]. At present, there is little evidence that transient hypercholesterolemia associated with anorexia nervosa leads to premature atherosclerotic cardiovascular disease, although structural cardiac complications of AN itself are well documented and long-term prospective data remain limited [17,19].

Consequently, elevated cholesterol in an underweight patient should be interpreted in the context of the overall clinical picture rather than considered an isolated indication for pharmacological treatment. The primary therapeutic goal should be correction of the underlying energy deficiency rather than normalisation of lipid concentrations alone.

5.2. Effect of nutritional rehabilitation on lipid profile

Nutritional rehabilitation remains the cornerstone of treatment for hypercholesterolemia associated with anorexia nervosa and other forms of severe undernutrition. Unlike primary dyslipidemia, in which cholesterol concentrations often require long-term pharmacological management, starvation-induced hypercholesterolemia usually improves as energy balance and body weight are restored [1].

Several longitudinal studies have demonstrated that total and LDL cholesterol concentrations generally decrease during nutritional rehabilitation, although the rate of normalisation varies considerably between individuals. In a prospective study of patients with anorexia nervosa, Rigaud et al. showed a biphasic lipoprotein pattern - low values at the very lowest body weights, normal values at an intermediate stage and elevated values as weight approached the higher end of malnourished range, with progressive normalisation of the lipid profile during refeeding [11]. This pattern indicated that the relationship between body weight and cholesterol is not linear; the most severely malnourished patients may show low rather than high cholesterol. Feillet et al., studying refeeding in adolescent girls with AN, found that endogenous cholesterol synthesis was relatively low at admission and rose significantly during 30 days of refeeding as apolipoprotein B fell and cholesterol level normalised. This observation

indicated that the initial elevation was not attributable to increased de novo cholesterol synthesis [20].

Importantly, persistent hypercholesterolemia after successful nutritional rehabilitation should be evaluated in terms of alternative causes of dyslipidemia, including familial hypercholesterolemia, hypothyroidism, chronic liver disease, or nephrotic syndrome. Therefore, serial assessment of lipid concentrations during recovery is often more informative than a single measurement obtained during acute starvation.

6. Clinical management of hypercholesterolemia in underweight individuals

6.1. Initial clinical assessment

The first step in managing hypercholesterolemia in an underweight patient is to identify the underlying cause of low body weight, which may include anorexia nervosa as defined by DSM-5 criteria [21]. A detailed medical history should include recent weight loss, eating habits, physical activity, gastrointestinal symptoms, medication use, and family history of premature cardiovascular disease or familial hypercholesterolemia [22]. In female patients, menstrual history may provide additional information about chronic energy deficiency and endocrine function [13].

Physical examination should assess nutritional status, body mass index (BMI), signs of malnutrition, and features suggesting alternative causes of dyslipidemia, such as hypothyroidism, chronic liver disease, nephrotic syndrome, or inherited lipid disorders. Because anorexia nervosa is frequently underdiagnosed, clinicians should also consider psychological assessment when restrictive eating behaviors or body image disturbance are suspected [23].

Routine laboratory evaluation should include a fasting lipid profile, complete blood count, liver function tests, renal function, thyroid-stimulating hormone (TSH), glucose, and inflammatory markers where clinically indicated. In selected patients, measurement of apolipoprotein B, lipoprotein(a), or genetic testing for familial hypercholesterolemia may be appropriate if lipid concentrations remain markedly elevated after nutritional recovery [24,25].

6.2. Is lipid-lowering therapy necessary?

Current evidence does not support routine lipid-lowering therapy in patients with starvation-related hypercholesterolemia. Unlike primary hypercholesterolemia, elevated cholesterol concentrations in anorexia nervosa are usually secondary to metabolic adaptations and often improve with nutritional rehabilitation [1,11]. Therefore, pharmacological treatment should

generally not be initiated during the acute phase of malnutrition unless another indication for lipid-lowering therapy is clearly present.

Instead, the primary therapeutic goal should be restoration of adequate energy intake and gradual weight recovery; refeeding should be managed carefully to avoid refeeding syndrome [26,27]. Nutritional rehabilitation not only improves body composition but also reverses many endocrine and metabolic adaptations responsible for impaired cholesterol metabolism. Serial monitoring of lipid concentrations during recovery is therefore recommended before considering statin therapy [11,22].

Persistent elevation of LDL cholesterol despite successful weight restoration should prompt reassessment for primary dyslipidemia or secondary causes unrelated to starvation. In such cases, referral to a lipid specialist may be appropriate.

6.3. Follow-up and monitoring

Patients should be monitored throughout nutritional rehabilitation to assess both clinical recovery and changes in lipid metabolism. Body weight, BMI, dietary intake, and lipid profile should be reassessed regularly. Because cholesterol concentrations may remain elevated for several weeks or months despite improved caloric intake, clinicians should interpret laboratory results together with the patient's nutritional progress rather than relying on isolated lipid measurements [11].

A repeat fasting lipid profile approximately 8–12 weeks after initiation of nutritional rehabilitation is reasonable in most patients. If substantial hypercholesterolemia persists after weight stabilization, additional investigations should be considered to exclude familial hypercholesterolemia, endocrine disorders, liver disease, or nephrotic syndrome [24].

Overall, management should focus on treating the underlying nutritional disorder rather than the biochemical abnormality itself. This approach minimizes unnecessary pharmacological treatment while ensuring that persistent dyslipidemia is appropriately investigated.

7. Practical recommendations

Based on the available evidence, hypercholesterolemia in underweight individuals should be interpreted in the context of nutritional status rather than as an isolated lipid abnormality. Since elevated cholesterol concentrations are often a reversible consequence of prolonged energy deficiency, management should primarily focus on identifying and treating the underlying cause of low body weight.

The following practical recommendations may be helpful in clinical practice:

- Assess nutritional status, recent weight loss, dietary habits, and possible eating disorders in every underweight patient presenting with hypercholesterolemia [22,23].
- Exclude secondary causes of dyslipidemia, including hypothyroidism, liver disease, nephrotic syndrome, and inherited lipid disorders when clinically indicated [24].
- Consider nutritional rehabilitation as the first-line treatment, as lipid abnormalities frequently improve with weight restoration [1,11].
- Avoid routine lipid-lowering therapy during acute malnutrition unless another clear indication for treatment exists [22].
- Repeat the lipid profile after nutritional recovery before establishing a diagnosis of persistent dyslipidemia or initiating long-term pharmacological therapy [11,24].
- Refer patients with persistent severe hypercholesterolemia despite adequate nutritional recovery for further specialist evaluation.

These recommendations emphasize that treatment should target the underlying nutritional disorder rather than cholesterol concentrations alone.

8. Limitations and future research directions

This review has several limitations. First, it is a narrative review and therefore does not include a formal risk-of-bias assessment or quantitative meta-analysis. Although the literature search was broad and focused on clinically relevant studies, it was not conducted according to systematic review methodology.

Another limitation is the relatively small amount of available evidence. Most studies investigating hypercholesterolemia in underweight individuals have been conducted in patients with anorexia nervosa, while data on other populations with chronic undernutrition, such as individuals with constitutional thinness [28], chronic illness, or Relative Energy Deficiency in Sport (RED-S) [29], remain limited. Many published studies are also based on relatively small sample sizes and differ in study design, diagnostic criteria, and duration of follow-up, making direct comparison difficult.

The mechanisms responsible for starvation-associated hypercholesterolemia have not yet been fully clarified. Although reduced LDL receptor activity, endocrine adaptations, altered cholesterol transport, and impaired bile acid metabolism have all been proposed, their relative contribution remains uncertain. In addition, long-term cardiovascular outcomes in patients with transient starvation-induced hypercholesterolemia have not been adequately investigated.

Future studies should include larger prospective cohorts and well-designed longitudinal studies evaluating lipid metabolism before, during, and after nutritional rehabilitation. Research

involving patients with RED-S, constitutional thinness, and other forms of chronic undernutrition would help determine whether the mechanisms described in anorexia nervosa can be generalized to other underweight populations. Further investigation is also needed to establish whether temporary hypercholesterolemia during starvation has any long-term clinical significance or represents a completely reversible metabolic adaptation.

9. Conclusions

Hypercholesterolemia in underweight individuals represents a well-recognized but often misunderstood metabolic phenomenon. Current evidence indicates that elevated cholesterol concentrations during prolonged energy deficiency are primarily the result of adaptive changes in cholesterol metabolism rather than excessive dietary cholesterol intake. Reduced LDL-receptor activity, endocrine adaptations, altered lipoprotein metabolism, and impaired cholesterol clearance appear to contribute to this process.

In most patients, hypercholesterolemia improves during nutritional rehabilitation, supporting the concept that treatment should primarily focus on restoring adequate nutritional status rather than routine lipid-lowering therapy. Careful clinical assessment remains essential to distinguish starvation-related hypercholesterolemia from primary or secondary disorders of lipid metabolism that require additional investigation or treatment.

Overall, the available evidence supports an individualized approach based on nutritional status, clinical presentation, and the underlying cause of low body weight. Although important advances have been made in understanding the metabolic adaptations associated with starvation, further research is needed to clarify the mechanisms responsible for hypercholesterolemia and to determine its long-term clinical implications in both patients with anorexia nervosa and other underweight populations.

Disclosure

Author contributions

Conceptualization: Julia Obersztyn, Aleksandra Mliczek, Adrianna Szczecińska

Methodology: Brajan Roczyński, Kinga Łompieś, Izabela Polok

Investigation: Julia Obersztyn, Aleksandra Mliczek, Izabela Polok, Kinga Łompieś

Supervision: Julia Obersztyn, Brajan Roczyński, Izabela Polok

Visualization: Aleksandra Mliczek, Adrianna Szczecińska, Kinga Łompieś

Writing - original draft: Julia Obersztyn, Aleksandra Mliczek, Adrianna Szczecińska

Writing - review and editing: Izabela Polok, Brajan Roczyński

Project administration: Brajan Roczyński, Kinga Łompiś, Adrianna Szczecińska

All authors have read and agreed to the published version of the manuscript.

Funding Statement: This research received no external funding.

Institutional Review Board Statement: Not applicable

Informed Consent Statement: Not applicable

Data Availability Statement: Data sharing is not applicable to this article.

Acknowledgments: Not applicable

Conflict of Interest Statement: The authors declare no conflicts of interest.

Declaration of the Use of Generative AI and AI-Assisted Technologies in the Writing Process:

During the preparation of this review, the authors used ChatGPT and Claude to improve language, grammar, text formatting and for bibliography verification. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

1. Hussain AA, Hübel C, Hindborg M, Lindkvist E, Kastrup AM, Yilmaz Z, et al. Increased lipid and lipoprotein concentrations in anorexia nervosa: a systematic review and meta-analysis. *Int J Eat Disord.* 2019 Jun;52(6):611–29. <https://doi.org/10.1002/eat.23051>
2. Gizzi G, Cataldi S, Mazzeschi C, Delvecchio E, Ceccarini MR, Codini M, et al. Hypercholesterolemia in Cancer and in Anorexia Nervosa: A Hypothesis for a Crosstalk. *Int J Mol Sci.* 2022 Jul 5;23(13):7466. <https://doi.org/10.3390/ijms23137466>
3. Winston AP. The clinical biochemistry of anorexia nervosa. *Ann Clin Biochem.* 2012 Mar 1;49(2):132–43. <https://doi.org/10.1258/acb.2011.011185>
4. Ikonen E. Cellular cholesterol trafficking and compartmentalization. *Nat Rev Mol Cell Biol.* 2008 Feb;9(2):125–38. <https://doi.org/10.1038/nrm2336>
5. Goldstein JL, Brown MS. Regulation of the mevalonate pathway. *Nature.* 1990 Feb 1;343(6257):425–30. <https://doi.org/10.1038/343425a0>

6. Cost J, Krantz MJ, Mehler PS. Medical complications of anorexia nervosa. *Cleve Clin J Med*. 2020 Jun;87(6):361–6. <https://doi.org/10.3949/ccjm.87a.19084>
7. Iwan K, Bielak A, Gryta J, Janczewska M, Kalicka M, Kolasa A, et al. How do eating disorders can affect the endocrine system? - literature review. *J Educ Health Sport*. 2023 Jun 5;37(1):140–52. <https://doi.org/10.12775/JEHS.2023.37.01.011>
8. Klop B, Elte JWF, Cabezas MC. Dyslipidemia in obesity: mechanisms and potential targets. *Nutrients*. 2013 Apr 12;5(4):1218–40. <https://doi.org/10.3390/nu5041218>
9. Brown MS, Goldstein JL. A Receptor-Mediated Pathway for Cholesterol Homeostasis. *Science*. 1986 Apr 4;232(4746):34–47. <https://doi.org/10.1126/science.3513311>
10. Weinbrenner T, Züger M, Jacoby G, Herpertz S, Liedtke R, Sudhop T, et al. Lipoprotein metabolism in patients with anorexia nervosa: a case–control study investigating the mechanisms leading to hypercholesterolaemia. *Br J Nutr*. 2004;91(6):959–969. <https://doi.org/10.1079/BJN20041151>
11. Rigaud D, Tallonneau I, Vergès B. Hypercholesterolaemia in anorexia nervosa: frequency and changes during refeeding. *Diabetes Metab*. 2009 Feb;35(1):57–63. <https://doi.org/10.1016/j.diabet.2008.08.004>
12. Ohwada R, Hotta M, Oikawa S, Takano K. Etiology of hypercholesterolemia in patients with anorexia nervosa. *Int J Eat Disord*. 2006 Nov;39(7):598–601. <https://doi.org/10.1002/eat.20298>
13. Westmoreland P, Krantz MJ, Mehler PS. Medical Complications of Anorexia Nervosa and Bulimia. *Am J Med*. 2016 Jan;129(1):30–7. <https://doi.org/10.1016/j.amjmed.2015.06.031>
14. Miller KK. Endocrine effects of anorexia nervosa. *Endocrinol Metab Clin North Am*. 2013 Sep;42(3):515–28. <https://doi.org/10.1016/j.ecl.2013.05.007>
15. NESTEL PJ. Cholesterol Metabolism in Anorexia Nervosa and Hypercholesterolemia. *J Clin Endocrinol Metab*. 1974 Feb 1;38(2):325–8. <https://doi.org/10.1210/jcem-38-2-325>
16. Gooding HC, de Ferranti SD. Cardiovascular risk assessment and cholesterol management in adolescents: getting to the heart of the matter. *Curr Opin Pediatr*. 2010 Aug;22(4):398–404. <https://doi.org/10.1097/MOP.0b013e32833a6e22>
17. Giovinazzo S, Sukkar SG, Rosa GM, Zappi A, Bezante GP, Balbi M, et al. Anorexia nervosa and heart disease: a systematic review. *Eat Weight Disord EWD*. 2019 Apr;24(2):199–207. <https://doi.org/10.1007/s40519-018-0567-1>
18. Mazurek K, Gęborys A, Wojcieszak M, Kania J, Miazga D, Maziarczyk A, et al. Systemic Consequences of Anorexia Nervosa: Multiorgan Disfunction and Long-Term

- Complications. Qual Sport. 2026 Feb 26;52:68741. <https://doi.org/10.12775/QS.2026.52.68741>
19. Buszek S, Widawska K, Bajak M, Czerniak P, Buszek J, Wajdowicz H, et al. Complications of anorexia nervosa - literature review. Qual Sport. 2024 Dec 13;35:56393. <https://doi.org/10.12775/QS.2024.35.56393>
 20. Feillet F, Feillet-Coudray C, Bard JM, Parra HJ, Favre E, Kabuth B, et al. Plasma cholesterol and endogenous cholesterol synthesis during refeeding in anorexia nervosa. Clin Chim Acta. 2000 Apr 1;294(1):45–56. [https://doi.org/10.1016/S0009-8981\(99\)00256-9](https://doi.org/10.1016/S0009-8981(99)00256-9)
 21. Diagnostic and statistical manual of mental disorders: DSM-5™, 5th ed. Arlington, VA, US: American Psychiatric Publishing, Inc.; 2013. xliv, 947 p. (Diagnostic and statistical manual of mental disorders: DSM-5™, 5th ed). <https://doi.org/10.1176/appi.books.9780890425596>
 22. Mehler PS, Brown C. Anorexia nervosa – medical complications. J Eat Disord. 2015 Mar 31;3(1):11. <https://doi.org/10.1186/s40337-015-0040-8>
 23. Morgan JF, Reid F, Lacey JH. The SCOFF questionnaire: assessment of a new screening tool for eating disorders. BMJ. 1999 Dec 4;319(7223):1467–8. <https://doi.org/10.1136/bmj.319.7223.1467>
 24. Grundy SM, Stone NJ, Bailey AL, Beam C, Birtcher KK, Blumenthal RS, et al. 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. J Am Coll Cardiol. 2019 Jun 25;73(24):e285–350. <https://doi.org/10.1016/j.jacc.2018.11.003>
 25. Sturm AC, Knowles JW, Gidding SS, Ahmad ZS, Ahmed CD, Ballantyne CM, et al. Clinical Genetic Testing for Familial Hypercholesterolemia: JACC Scientific Expert Panel. J Am Coll Cardiol. 2018 Aug 7;72(6):662–80. <https://doi.org/10.1016/j.jacc.2018.05.044>
 26. Mehanna HM, Moledina J, Travis J. Refeeding syndrome: what it is, and how to prevent and treat it. BMJ. 2008 Jun 28;336(7659):1495–8. <https://doi.org/10.1136/bmj.a301>
 27. Marko N, Niewiadomska J, Furtak D, Grela W, Tulej D, Głogowska P, et al. Refeeding syndrome as a serious complication of a treatment of patients with anorexia nervosa - a literature review. J Educ Health Sport. 2025 Apr 3;80:58488. <https://doi.org/10.12775/JEHS.2025.80.58488>

28. Estour B, Marouani N, Sigaud T, Lang F, Fakra E, Ling Y, et al. Differentiating constitutional thinness from anorexia nervosa in DSM 5 era. *Psychoneuroendocrinology*. 2017 Oct;84:94–100. <https://doi.org/10.1016/j.psyneuen.2017.06.015>
29. Mountjoy M, Sundgot-Borgen J, Burke L, Carter S, Constantini N, Lebrun C, et al. The IOC consensus statement: beyond the Female Athlete Triad—Relative Energy Deficiency in Sport (RED-S). *Br J Sports Med*. 2014 Apr;48(7):491–7. <https://doi.org/10.1136/bjsports-2014-093502>